

Github: https://github.com/ReneStander/BMG_R_Python_2026

Introduction to Python Programming

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Department of Statistics, University of Pretoria

4 & 6 May 2026

BILL & MELINDA
GATES *foundation*

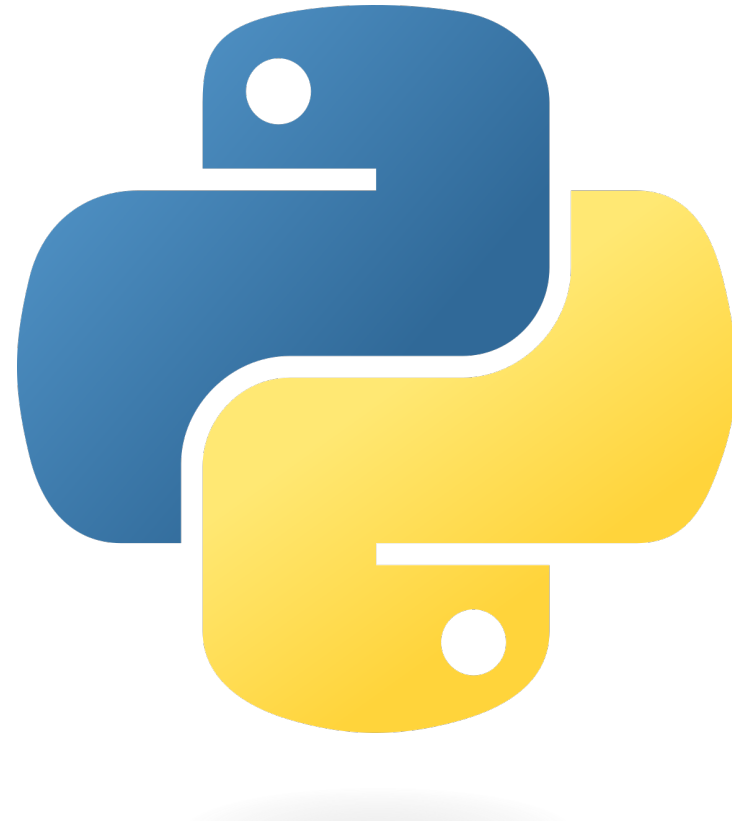


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Outline

1. Introduction to Python
2. Basics of the programming language
3. Data manipulation
4. Data visualization
5. Basic statistical analysis



Program

09:00 – 10:30	Session 1
10:30 – 10:45	Break
10:45 – 12:00	Session 2
12:00 – 13:00	Lunch
13:00 – 14:30	Session 3
14:30 – 14:45	Break
14:45 – 16:00	Session 4

Resources

- Downey, A.B., 2012. *Think python*. O'Reilly Media, Inc.
<https://alldowney.github.io/ThinkPython/#>
- Adhikari, A., DeNero, J. and Wagner, D., 2022. *Computational and Inferential Thinking: The Foundations of Data Science*, Second edition, University of California, Berkeley.
<https://inferentialthinking.com/>

Introduction to Python

Why Python?

- Mature programming language.
- Has excellent properties for newbie programmers.
- Currently one of the most flexible programming languages.
- Has a large ecosystem of libraries making it easy for Data Scientists.



Applications developed with Python you may know...



NETFLIX



Running Python in Google Colab

Google Colab

- Hosted **Jupyter Notebook** service.
- Required **no** hardware or software to be installed on your device.
- Provides free access to computing resources such as GPUs.

<https://colab.google/notebooks/>

Running Python in Google Colab

The screenshot shows the Google Colab web interface. At the top, the file is named "Untitled1.ipynb" with a star icon next to it. A green box labeled "Name your file" has an arrow pointing to the file name. Below the file name is a menu bar with options: File, Edit, View, Insert, Runtime, Tools, Help, and a link for "All changes saved". On the right side of the top bar, there are icons for "Comment", "Share", a settings gear, and a user profile picture. A green box labeled "You can share the file with your collaborators" has an arrow pointing to the "Share" icon. In the center, there is a toolbar with "+ Code" and "+ Text" buttons. A green box labeled "You can add code chunks and text boxes" has an arrow pointing to these buttons. Below the toolbar is a code editor area with a prompt: "Start coding or generate with AI." On the right side of the editor, there is a "Connect" dropdown menu and a "Gemini" button. Below these are several icons for file operations: upload, download, link, comment, settings, print, delete, and a more options menu. A green box labeled "You can share the file with your collaborators" also has an arrow pointing to the "Share" icon in the top right corner.

Untitled1.ipynb ☆

File Edit View Insert Runtime Tools Help [All changes saved](#)

+ Code + Text

Start coding or generate with AI.

Comment Share ⚙️ 👤

Connect Gemini

↑ ↓ 🔗 💬 ⚙️ 🖨️ 🗑️ ⋮

Name your file

You can add code chunks and text boxes

You can share the file with your collaborators

Running Python in Google Colab

The image shows the Google Colab interface. At the top, the file name is 'Untitled1.ipynb'. The menu bar includes File, Edit, View, Insert, Runtime, Tools, and Help. The 'Runtime' menu is open, displaying various execution options with their respective keyboard shortcuts. The 'Change runtime type' option is highlighted with a green box. To the right of the menu, the status bar shows 'Connected' and 'Gemini'. Below the menu, a dialog box titled 'Change runtime type' is open. It features a 'Runtime type' dropdown menu set to 'Python 3', which is also highlighted with a green box. Below this, the 'Hardware accelerator' section shows five options: CPU (selected with a blue radio button), T4 GPU, A100 GPU, L4 GPU, and TPU v2-8. At the bottom of the dialog, there is a link to 'Purchase additional compute units' and buttons for 'Cancel' and 'Save'.

Untitled1.ipynb ☆

File Edit View Insert Runtime Tools Help All changes saved

+ Code + Text

Start coding or

Run all Ctrl+F9

Run before Ctrl+F8

Run the focused cell Ctrl+Enter

Run selection Ctrl+Shift+Enter

Run after Ctrl+F10

Interrupt execution Ctrl+M I

Restart session Ctrl+M .

Restart session and run all

Disconnect and delete runtime

Change runtime type

Manage sessions

View resources

View runtime logs

Comment Share

Connected Gemini

Change runtime type

Runtime type

Python 3

Hardware accelerator ?

☒ CPU ☐ T4 GPU ☐ A100 GPU ☐ L4 GPU

☐ TPU v2-8

Want access to premium GPUs? [Purchase additional compute units](#)

Cancel Save

Running Python in Google Colab

The screenshot displays the Google Colab interface. On the left, the 'Files' sidebar is open, showing a file explorer with a folder named 'sample_data'. A green box labeled '1' highlights the folder icon, and a green box labeled '2' highlights the upload icon. A green box labeled '3' points to a text box that says 'Navigate the file explorer that pops up to select the file that you want to import'. The main area shows a code editor with a prompt 'Start coding or generate with AI.' and a toolbar with various icons. The bottom status bar indicates 'Connected to Python 3 Google Compute Engine backend' and shows a disk usage bar with '74.89 GB available'.

1

2

3

Navigate the file explorer that pops up to select the file that you want to import

Files

sample_data

+ Code + Text

Start coding or generate with AI.

RAM

Disk

Gemini

<>

Disk

74.89 GB available

Connected to Python 3 Google Compute Engine backend

Installing Python

1. Download Anaconda navigator. <https://www.anaconda.com/download>



2. Access the Python IDE, **Spyder**, through the Anaconda navigator.



Fundamental Python Libraries for Data Scientists

- **NumPy:** Provides support for multidimensional **arrays with basic operations on them** and useful linear algebra functions.
- **SciPy:** Provides a collection of **numerical algorithms** and domain-specific toolboxes, including signal processing, optimization and statistics.
- **Matplotlib:** Enables **data visualization**.
- **Pandas:** Provides high performance **data structures** and data analysis tools.
- **Scikit-Learn:** Offers simple and efficient tools for **common tasks in data analysis** such as classification, regression, clustering, and many more...

Coding scripts for this course



BMG_Intro_R_Python Public



Pin



Watch



main ▾



1 Branch



0 Tags



Go to file



<> Code ▾



ReneStander Created using Colab

6e7e9c9 · 3 minutes ago



11 Commits



Intro to Python

Created using Colab

3 minutes ago



Intro to R

Prepare files for BMG 2025

13 hours ago

Coding scripts for this course

The screenshot shows a GitHub repository interface. The top navigation bar includes links for Code, Issues, Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings. The left sidebar shows the file explorer with a tree view containing folders 'Intro to Python' and 'Intro to R'. The 'Intro to Python' folder is expanded, showing several files, with 'BMG Workshop - Intro + Basics....' selected. The main content area displays the selected file, 'BMG_Intro_R_Python / Intro to Python / BMG Workshop - Intro + Basics.ipynb'. It shows the file was created by 'ReneStander' using Colab, 9aa7180, 9 minutes ago. Below this, there are tabs for 'Preview', 'Code', and 'Blame'. The 'Preview' tab is active, showing a preview of the Jupyter Notebook. A green box highlights the 'Open in Colab' button. The notebook content includes the title 'Introduction to Python workshop', the author 'Dr Rene Stander', the date '30 September 2025', and a list of sources.

<> Code Issues Pull requests Actions Projects Wiki Security Insights Settings

Files

main

Go to file

Intro to Python

- BMG Workshop - Basic statistic...
- BMG Workshop - Basic statistic...
- BMG Workshop - Data manipul...
- BMG Workshop - Data visualisa...
- BMG Workshop - Intro + Basics....**
- RacingGameData.csv
- Spiders.csv
- SuperAnimals.csv

Intro to R

BMG_Intro_R_Python / Intro to Python / BMG Workshop - Intro + Basics.ipynb

ReneStander Created using Colab 9aa7180 · 9 minutes ago History

Preview Code Blame

Raw Copy Download Edit

Open in Colab

Introduction to Python workshop

Dr Rene Stander

30 September 2025

Sources:

- Downey, AB., 2012. Think Python, O'Reilly Media, Inc. (<https://alldowney.github.io/ThinkPython/>)
- Adhikari, A., DeNero, J. and Wagner, D., 2022. Computational and Inferential Thinking: The Foundations of Data Science, Second edition, University of California, Berkeley. (<https://inferentialthinking.com/chapters/intro.html>)

Coding scripts for this course

The screenshot displays a JupyterLab environment. At the top, the title bar reads 'BMG Workshop - Intro + Basics.ipynb'. Below it is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A toolbar contains 'Commands', '+ Code', '+ Text', 'Run all', and a 'Copy to Drive' button. On the right, there are 'Share' and 'Connect' buttons, and a user profile icon. A left sidebar shows icons for a menu, search, code, key, and file explorer. The main area contains a notebook with the title 'Introduction to Python workshop' and author 'Dr Rene Stander', dated '30 September 2025'. It lists two sources: 'Downey, AB., 2012. Think Python, O'Reilly Media, Inc.' and 'Adhikari, A., DeNero, J. and Wagner, D., 2022. Computational and Inferential Thinking: The Foundations of Data Science, Second edition, University of California, Berkeley.' Below this, a section titled 'Introduction to Python' includes the instruction 'Check the Python version and the installed packages'. At the bottom, a code cell is partially visible with the comment '# Python version'.

co BMG Workshop - Intro + Basics.ipynb
File Edit View Insert Runtime Tools Help

Q Commands | + Code + Text | ▶ Run all ▼ Copy to Drive

Connect

Introduction to Python workshop

Dr Rene Stander

30 September 2025

Sources:

- Downey, AB., 2012. Think Python, O'Reilly Media, Inc. (<https://alldowney.github.io/ThinkPython/>)
- Adhikari, A., DeNero, J. and Wagner, D., 2022. Computational and Inferential Thinking: The Foundations of Data Science, Second edition, University of California, Berkeley. (<https://inferentialthinking.com/chapters/intro.html>)

▼ Introduction to Python

Check the Python version and the installed packages

[]

```
# Python version
```

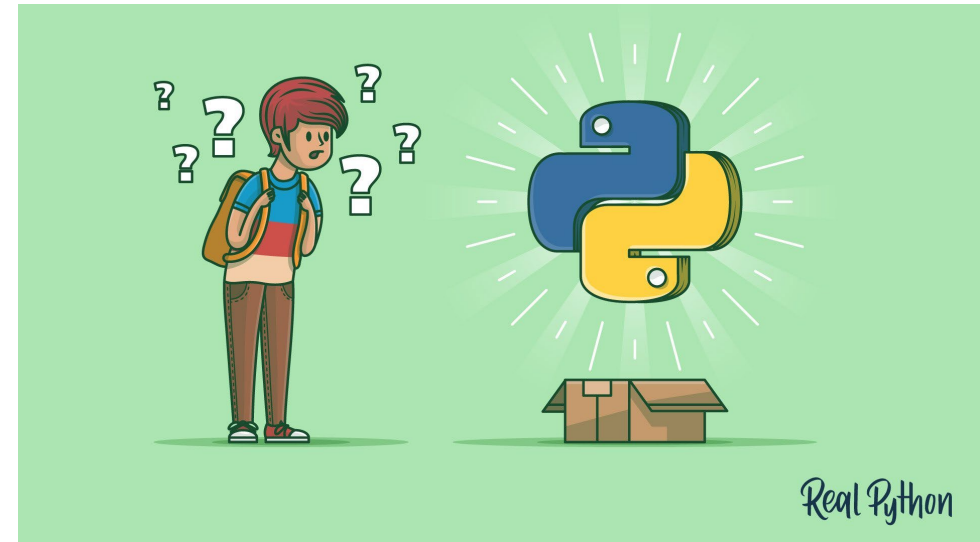

Intro + Basics

Open the notebook named **BMG Workshop - Intro + Basics.ipynb** in Google Colab

This part of the work does not require any additional data sets

Github:

https://github.com/ReneStander/BMG_R_Python_2026



Data manipulation

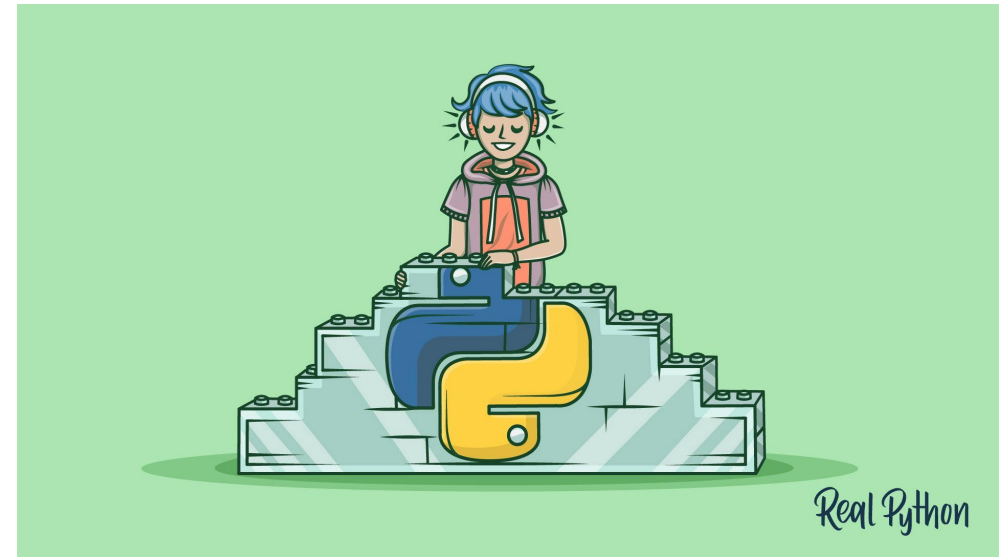
Open the notebook named **BMG Workshop – Data manipulation.ipynb** in Google Colab

To run the code in this notebook, please upload the following data sets into Google Colab:

- **RacingGameData.csv**

Github:

https://github.com/ReneStander/BMG_R_Python_2026



Data visualisation

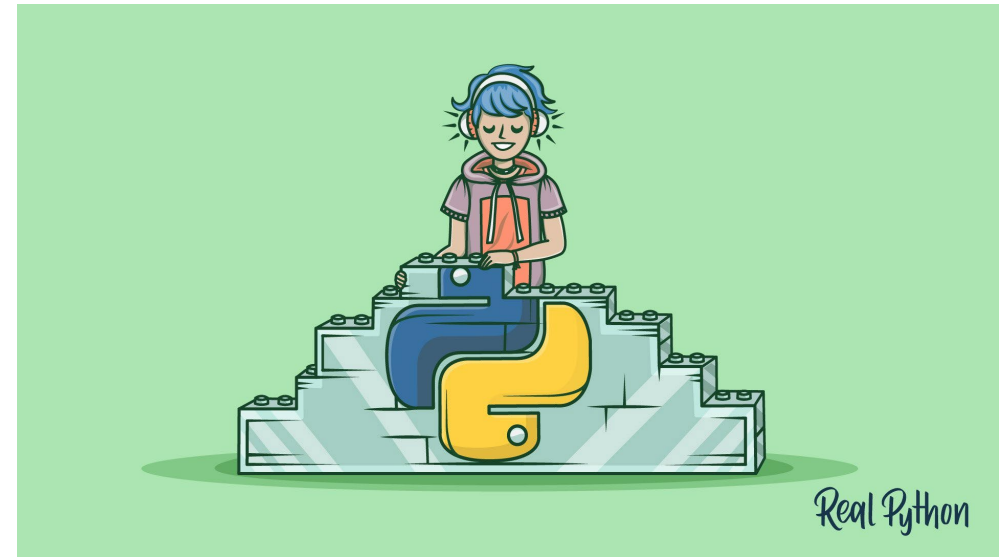
Open the notebook named **BMG Workshop – Data visualisation.ipynb** in Google Colab

To run the code in this notebook, please upload the following data sets into Google Colab:

- **SuperAnimals.csv**
- **Spiders.csv**

Github:

https://github.com/ReneStander/BMG_R_Python_2026



Data visualisation

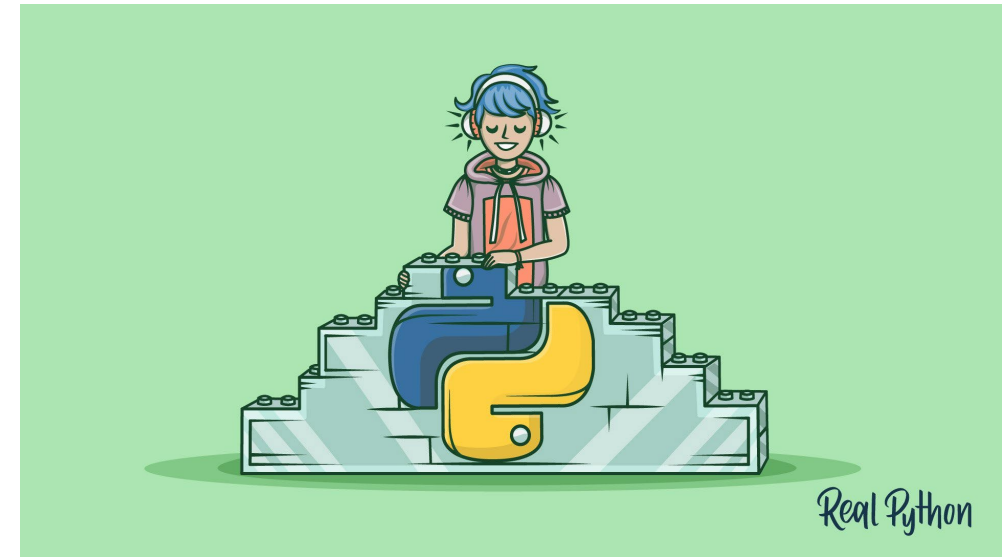
Open the notebook named **BMG Workshop – Data visualisation.ipynb** in Google Colab

To run the code in this notebook, please upload the following data sets into Google Colab:

- **SuperAnimals.csv**
- **Spiders.csv**

Github:

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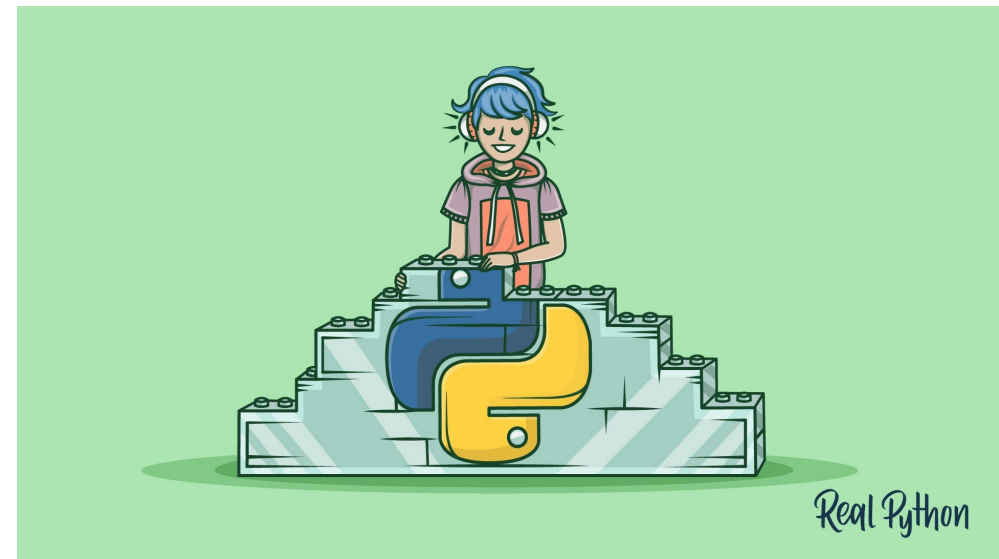
Statistical analysis – Categorical variables

Open the notebook named **BMG Workshop – Basic statistical analysis – Categorical variables.ipynb** in Google Colab

This part of the work does not require any additional data sets

Github:

https://github.com/ReneStander/BMG_R_Python_2026



Statistical analysis – Categorical variables

Open the notebook named **BMG Workshop – Basic statistical analysis – Numerical variables.ipynb** in Google Colab

To run the code in this notebook, please upload the following data sets into Google Colab:

- **SuperAnimals.csv**
- **Spiders.csv**

Github:

https://github.com/ReneStander/BMG_R_Python_2026

